Qualification Pack

Solar Photovoltaic Entrepreneur

QP Code: SGJ/Q0901

Version: 3.0

NSQF Level: 4

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Contents

SGJ/Q0901: Solar Photovoltaic Entrepreneur

Brief Job Description

Solar Photovoltaic Entrepreneur is an individual who ventures into Solar market to lead an enterprise. He/She has the understanding of solar business models, market, technical knowledge of rooftop solar PV plants, along with components procurement and financing. Solar Photovoltaic Entrepreneur can prepare the feasibility study report and perform basic energy generation forecasting using simulation software and is also responsible for the managing the complete Solar PV project lifecycle.

Personal Attributes

The individual shall have an ability to take initiative, handle pressure and shall be motivated with good interpersonal and problem solving skills. The individual shall have leadership skills and organized with their work and act with integrity while performing multiple tasks for the customers with quality and timely deliverables.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. SGJ/N0111: Set up New Venture for Solar Photovoltaic System
2. SGJ/N4125: Business Model and site feasibility study report
3. SGJ/N0148: Manage Installation of Solar Power Plant
4. SGJ/N4126: Operation and maintenance of Solar Rooftop, Ground Mounted PV Plant
5. SGJ/N0106: Maintain Personal Health & Safety at project site
6. DGT/VSQ/N0102: Employability Skills (60 Hours)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
Country	India
NSQF Level	4

Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2431.0501
Minimum Educational Qualification & Experience	12th grade Pass OR 10th Class with 3 Years of experience in relevant field Renewable energy/Solar PV Sector OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience in renewable Energy/Solar PV Sector OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Renewable energy/Solar Pv sector
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-04-ES-04052-2025-V2-SCGJ
NQR Version	3

Remarks:

Total 480 hours: 225 Hours of Theory with Employability Skills, 225 Hours of Practical, 30 Hours of OJT

SGJ/N0111: Set up New Venture for Solar Photovoltaic System

Description

This unit provides an overview of Solar PV Sector with evolving business opportunities in terms of technology, finance & regulatory landscape of solar energy. This unit also provides a brief understanding about developing entrepreneurship skills for starting a new venture and managing it.

Scope

The scope covers the following :

- Introduction to Solar PV sector in India
- Set up new venture for Solar Photovoltaic

Elements and Performance Criteria

Introduction to Solar PV Sector in India

To be competent, the user/individual on the job must be able to:

PC1. discuss and provide overview of solar PV technology, rooftop and ground mounted solar sector in India

PC2. discuss various market research reports and industrial magazines present in the market

PC3. discuss different type of ground mounted and Rooftop Solar PV Power Plants and their working principles

PC4. discuss and provide an overview of grid interactive and off- grid Solar Sector in India

PC5. discuss type of off- grid Solar PV Power devices and their working principles

PC6. discuss various components of a solar PV system and their operating principles

PC7. Explain the broader business, finance, technology & regulatory landscape of solar energy in the country

PC8. discuss solar energy and power sector landscape in the country

PC9. discuss benefits of solar energy over conventional sources of energy

PC10. discuss the broader business, finance, technology & regulatory landscape of solar energy in the country

PC11. discuss how to analyse the basics of electrical concepts like voltage, current, power, energy, etc

PC12. discuss typical specifications, functioning, operating principle, maintenance requirements, handling procedures and warranties of different types of solar PV plant components

PC13. discuss how to interpret sample Electricity bill of a customer, notices and/or cautions

PC14. discuss how to identify various financial institutions and banks involved in lending for solar power projects and analyse their terms & conditions associated with loans

PC15. discuss emerging solar entrepreneurship landscape including success stories of various start-ups

Set up new venture for Solar Photovoltaic

To be competent, the user/individual on the job must be able to:

PC16. discuss the process for setting up a new venture including required registrations and availing low-cost financing from Financing Institutions (FIs) and Venture Capital (VC) firms
PC17. discuss various evolving opportunities across solar solutions, services and market ecosystem
PC18. discuss how to identify the key ingredients of a business plan
PC19. discuss how to distinguish between fixed and working capital requirements
PC20. discuss the components of a loan application
PC21. Discuss the risks associated with different ventures
PC22. discuss how to demonstrate leadership skills and effective resource management techniques
PC23. discuss the use of MS Word and Excel for preparing a business proposal
PC24. discuss to prepare a workable presentation for marketing and business development
PC25. discuss how to choose the right buyer in a given situation of market parameters
PC26. discuss how to identify the challenges and risks for new entrepreneurs
PC27. discuss various taxation and related compliances to set up and operate a solar business
PC28. discuss the key aspects to successfully run a solar business
PC29. Explain how to interact and convince with customers for the Solar Photovoltaic installation
PC30. discuss how to evolving Start- ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio is improving overall solar landscape
PC31. Discuss the significance of following the recommended gender-inclusive practices in hiring and work management
PC32. Highlight the ways of creating a conducive environment at the enterprise for all genders
PC33. List ways to make gender diversity and inclusion a company initiative and priority
PC34. show how to the process for setting up a new venture including registrations and availing financing from FIs and VCs
PC35. show how to identify the key ingredients of a business plan
PC36. show the process of a loan application
PC37. show the application of tools including MS Word and Excel for preparing a business proposal
PC38. Demonstrate the importance of leadership skills and effective resource management techniques.
PC39. Demonstrate how to choose the right buyer in a given situation of market parameters.
PC40. show how to prepare a workable presentation for marketing and business development
PC41. Show how to interact and convince with customers for the Solar Photovoltaic installation
PC42. show how to choose the right buyer in a given situation of market parameters
PC43. show how to identify the challenges and risks for new entrepreneurs
PC44. show how to the importance of leadership skills and effective resource management techniques
PC45. show the key aspects to successfully run a solar business
PC46. show how evolving Start- ups across solar Consultancy/O&M/Installation and EPC, and diversifying solar portfolio is improving overall solar landscape
PC47. show how to adhere to various taxation and related compliances to set up and operate a solar business
PC48. Role-play a situation on how to interact with personnel, clients, and vendors using gender-neutral statements.

PC49. Role play a situation on how to train an employee, irrespective of gender

PC50. Role-play a situation on how to create a culture of shared accountability

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. government/corporate policies and guidelines on solar pv, solar rooftop

KU2. companys work safety policy

KU3. company's customer support policy.

KU4. companys documentation policy.

KU5. obtain authorization from specified field safety officer and supervisor.

KU6. companys different department.

KU7. definition of entrepreneurship from different perspectives

KU8. outline the importance of entrepreneurship: enhances creativity and innovation, builds self confidence in people, serves as a tool for nation building, serves as the engine of growth for the nations economy.

KU9. explain the reasons why entrepreneurship should be developed in a country: reasons include: employment generation, increased national production and re-investing national resources

KU10. state the characteristics of an entrepreneur: characteristics of the entrepreneurs, risk taking, innovation and creativity, opportunity orientation

KU11. explain the challenges/problems facing small businesses like financing and access to markets, government policies and inadequate managerial skills

KU12. describe the procedure for registering a business by defining a business idea, source of business idea, programs/ procedure and available schemes.

KU13. state the process of starting a new enterprises process by mobilizing and reorganizing resources.

KU14. study of different pictorial expression of non-verbal communication and its analysis

KU15. time management concepts including discipline, punctuality, act in time on commitment and quality productive time

KU16. ability to shape and direct working/process methods according to self-defined criteria

KU17. empathize: comprehend other opinions points of views, and face them with understanding

KU18. learn ms word and ms excel: creating, organizing & formatting content, collaborating merge, insert, view, edit, track mode etc.

KU19. understand the fixed and capital working requirements for running a business

KU20. understand how to make a business plan

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. prepare and maintain documentation

GS2. read vernacular/English language

GS3. read and understand manuals, health and safety instructions, memos, other company documents

GS4. read from different sources- books, screens in machines and signage

GS5. read various colour codes, as per standard electrical, mechanical and civil nomenclature

GS6. express statements or information clearly so that others can hear and understand

GS7. participate in and understand the main points of simple discussions

GS8. respond appropriately to any queries

GS9. communicate with employees.

GS10. define organization rule- based decision making process

GS11. take decision with systematic course of actions and/or response

GS12. plan and organize work schedule to meet deadlines

GS13. work constructively and collaboratively with others

GS14. prepare organization code of conduct

GS15. manage relationships with customers with intent on satisfying its requirements for service delivery

GS16. recognize problems and search for solutions

GS17. choose best methods to complete assigned tasks

GS18. apply domain knowledge, observations and data to select course of action to perform tasks related to solar photovoltaic power plant.

GS19. critically evaluate information obtained from customers and workers to perform day to day activities

GS20. ask questions for better understanding

GS21. practice and acceptance of gender and its concepts

GS22. empathy across genders and towards PwD

GS23. communicate with team members and colleagues on the significance of greening of jobs

GS24. make timely decisions for efficient utilization of resources

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Solar PV Sector in India	**52**	**-**	**-**	**-**
PC1. discuss and provide overview of solar PV technology, rooftop and ground mounted solar sector in India	3	-	-	-
PC2. discuss various market research reports and industrial magazines present in the market	3	-	-	-
PC3. discuss different type of ground mounted and Rooftop Solar PV Power Plants and their working principles	4	-	-	-
PC4. discuss and provide an overview of grid interactive and off- grid Solar Sector in India	5	-	-	-
PC5. discuss type of off- grid Solar PV Power devices and their working principles	6	-	-	-
PC6. discuss various components of a solar PV system and their operating principles	5	-	-	-
PC7. Explain the broader business, finance, technology & regulatory landscape of solar energy in the country	2	-	-	-
PC8. discuss solar energy and power sector landscape in the country	2	-	-	-
PC9. discuss benefits of solar energy over conventional sources of energy	2	-	-	-
PC10. discuss the broader business, finance, technology & regulatory landscape of solar energy in the country	2	-	-	-
PC11. discuss how to analyse the basics of electrical concepts like voltage, current, power, energy, etc	4	-	-	-
PC12. discuss typical specifications, functioning, operating principle, maintenance requirements, handling procedures and warranties of different types of solar PV plant components	3	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. discuss how to interpret sample Electricity bill of a customer, notices and/or cautions	3	-	-	-
PC14. discuss how to identify various financial institutions and banks involved in lending for solar power projects and analyse their terms & conditions associated with loans	6	-	-	-
PC15. discuss emerging solar entrepreneurship landscape including success stories of various start-ups	2	-	-	-
Set up new venture for Solar Photovoltaic	**30**	**18**	**-**	**-**
PC16. discuss the process for setting up a new venture including required registrations and availing low-cost financing from Financing Institutions (FIs) and Venture Capital (VC) firms	2	-	-	-
PC17. discuss various evolving opportunities across solar solutions, services and market ecosystem	2	-	-	-
PC18. discuss how to identify the key ingredients of a business plan	1	-	-	-
PC19. discuss how to distinguish between fixed and working capital requirements	2	-	-	-
PC20. discuss the components of a loan application	2	-	-	-
PC21. Discuss the risks associated with different ventures	2	-	-	-
PC22. discuss how to demonstrate leadership skills and effective resource management techniques	2	-	-	-
PC23. discuss the use of MS Word and Excel for preparing a business proposal	1	-	-	-
PC24. discuss to prepare a workable presentation for marketing and business development	3	-	-	-
PC25. discuss how to choose the right buyer in a given situation of market parameters	1	-	-	-
PC26. discuss how to identify the challenges and risks for new entrepreneurs	3	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. discuss various taxation and related compliances to set up and operate a solar business	2	-	-	-
PC28. discuss the key aspects to successfully run a solar business	1	-	-	-
PC29. Explain how to interact and convince with customers for the Solar Photovoltaic installation	2	-	-	-
PC30. discuss how to evolving Start- ups across solar Consultancy/O&M/Installation and EPC space, and diversifying solar portfolio is improving overall solar landscape	1	-	-	-
PC31. Discuss the significance of following the recommended gender-inclusive practices in hiring and work management	1	-	-	-
PC32. Highlight the ways of creating a conducive environment at the enterprise for all genders	1	-	-	-
PC33. List ways to make gender diversity and inclusion a company initiative and priority	1	-	-	-
PC34. show how to the process for setting up a new venture including registrations and availing financing from FIs and VCs	-	1	-	-
PC35. show how to identify the key ingredients of a business plan	-	1	-	-
PC36. show the process of a loan application	-	1	-	-
PC37. show the application of tools including MS Word and Excel for preparing a business proposal	-	1	-	-
PC38. Demonstrate the importance of leadership skills and effective resource management techniques.	-	1	-	-
PC39. Demonstrate how to choose the right buyer in a given situation of market parameters.	-	1	-	-
PC40. show how to prepare a workable presentation for marketing and business development	-	1	-	-
PC41. Show how to interact and convince with customers for the Solar Photovoltaic installation	-	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC42. show how to choose the right buyer in a given situation of market parameters	-	1	-	-
PC43. show how to identify the challenges and risks for new entrepreneurs	-	1	-	-
PC44. show how to the importance of leadership skills and effective resource management techniques	-	1	-	-
PC45. show the key aspects to successfully run a solar business	-	1	-	-
PC46. show how evolving Start- ups across solar Consultancy/O&M/Installation and EPC, and diversifying solar portfolio is improving overall solar landscape	-	1	-	-
PC47. show how to adhere to various taxation and related compliances to set up and operate a solar business	-	1	-	-
PC48. Role-play a situation on how to interact with personnel, clients, and vendors using gender-neutral statements.	-	1	-	-
PC49. Role play a situation on how to train an employee, irrespective of gender	-	2	-	-
PC50. Role-play a situation on how to create a culture of shared accountability	-	1	-	-
NOS Total	**82**	**18**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0111
NOS Name	Set up New Venture for Solar Photovoltaic System
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Entrepreneur
NSQF Level	4
Credits	4
Version	5.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

SGJ/N4125: Business Model and site feasibility study report

Description

This covers the understanding of Indian solar industry landscape, business models, and policy framework driving rooftop solar adoption. It details site assessment, design considerations, load profiling, resource analysis, and system sizing for rooftop solar PV plants. Learners gain skills in feasibility study preparation, financial evaluation, and risk assessment using industry-standard tools like PVsyst and PV Sol.

Scope

The scope covers the following :

- This NOS covers the business, technical, and policy aspects of rooftop solar PV projects. It includes site assessment, system design, financial evaluation, and feasibility reporting using standard tools. The scope extends to risk assessment, stakeholder roles, and compliance with MNRE, state, and DISCOM regulations.

Elements and Performance Criteria

Business Models in Solar PV

To be competent, the user/individual on the job must be able to:

PC1. Discuss the overview of the Indian solar industry, key stakeholders (developers, EPCs, DISCOMs, financiers), and market drivers.
PC2. Discuss different types of business models CAPEX, OPEX/RESCO, Lease, BOOT, and Hybrid models
PC3. Explain the structure, ownership, financial responsibility, advantages, and typical users of capex model.
PC4. Discuss service-based approach, power purchase agreements (PPA), tariff structures, and stakeholder roles for OPEX model.
PC5. Business logic for shared solar, community solar systems, and consumer aggregation
PC6. Describe payback periods, IRR, net savings, and other performance indicators.
PC7. Discuss the role of MNRE, state policies, subsidies, and DISCOM regulations influencing business models.
PC8. Case study analysis of a rooftop solar project using CAPEX model
PC9. Design a basic PPA for a RESCO project
PC10. Calculate Simple IRR and payback period calculation using Excel
PC11. Customer pitch for different business models
PC12. Show how to do mapping of solar business models to different consumer categories (residential, C&I, government)
PC13. Show the role of MNRE, state policies, subsidies, and DISCOM regulations influencing business models by visiting their website.

Site Feasibility Study Rooftop Solar

To be competent, the user/individual on the job must be able to:

PC14. Identify optimum location of rooftop solar plants installations

PC15. Assess the site level pre-requisites for solar panel installation

PC16. Discuss how to decide on the type of mounting to be constructed and place of mounting as per client requirement

PC17. Discuss how to check for any shading obstacles

PC18. Discuss to prepare a site map of the location where installation has to be carried out.

PC19. Discuss how to assess the load to be run on solar PV power plant and prepare a load profile

PC20. Explain to estimate the capacity of solar PV power plant

PC21. Discuss on how to decide on battery backup as per grid availability, loads and client expectation

PC22. Assess or obtain the site-specific major parameters of solar resource data like GHI, DNI, Temperature and Wind

PC23. Explain how to perform shading analysis

PC24. Discuss how to estimate the energy generated from the rooftop solar PV power plant using solar design software like PV Sol, PVsyst, etc.

PC25. Identify the risks associated with the specific solar project

PC26. Explain how to prepare a site Feasibility Study Report using specialized software like PV sol, PV Syst, etc.

PC27. Demonstrate how to identify optimum location of rooftop solar plants installations

PC28. Demonstrate how to assess the site level pre-requisites for solar panel installation.

PC29. Show how to decide on the type of mounting to be constructed and place of mounting as per client requirement.

PC30. Demonstrate how to check for any shading obstacles, perform shading analysis, and to prepare a site map of the location where installation has to be carried out.

PC31. Analyze how to assess the load to be run on solar PV power plant, prepare a load profile and estimate the capacity of solar PV power plant

PC32. Demonstrate how to decide on battery backup as per grid availability, loads and client expectation

PC33. Show how to obtain the site-specific parameters of solar resource data like GHI, DNI, Temperature and Wind

PC34. Show how to estimate the energy generated from the rooftop solar PV power plant using solar design software like PV SOL, PVsyst, etc.

PC35. Demonstrate the risks associated with the specific solar project and mitigate those

PC36. Demonstrate how to support a site Feasibility Study Report using specialized software like PVsyst, etc

Site Feasibility Study of Ground Mounted Solar

To be competent, the user/individual on the job must be able to:

PC37. Explain how to assess irradiation and other geological data

PC38. Explain how to assess the daily, monthly and annual solar resource data including GHI, ONI, Albedo etc. for site to evaluate the potential for solar energy generation at the site.

PC39. Explain how to collect data of local weather conditions such as temperature range, flooding, wind speed, humidity, pollution levels, snow and other climatic conditions for assessment of its impact on solar energy generation.

PC40. Explain how to conduct the soil testing and contour mapping.

PC41. Explain how to conduct the detailed survey plan of the land proposed for installation of solar power plant with elevations and topography contour mapping etc.

PC42. Discuss how to assess physical site accessibility

PC43. Discuss to identify the policy, regulations and procedures for ground mounted solar sector.

PC44. Discuss how to assess transmission line & power evacuation

PC45. Demonstrate how to retrieve irradiation data from site

PC46. Analyse daily, monthly and annual resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site.

PC47. Analyse the soil test report

PC48. Demonstrate how to use site survey tools like, hand held irradiation meter, compass.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Indian solar industry landscape, key stakeholders (developers, EPCs, DISCOMs, financiers), business models (CAPEX, OPEX/RESCO, Lease, BOOT, Hybrid), and policy/regulatory framework.

KU2. Explain financial aspects of solar projects including payback period, IRR, NPV, tariffs, and savings with reference to consumer categories (residential, C&I, government).

KU3. Assess site prerequisites for rooftop/ground-mounted solar including load profile, shading, orientation, roof/land suitability, and solar resource data (GHI, DNI, temperature, wind)

KU4. Demonstrate the use of solar design and analysis tools (PVsyst, PVSOL, shading meters) for energy estimation, feasibility reports, and risk assessment.

KU5. Evaluate the role of policies, subsidies, DISCOM regulations, and PPAs in shaping solar business models and consumer adoption.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate how to retrieve, analyze, and interpret solar resource data using site instruments (irradiation meter, compass, GPS) and software tools.

GS2. Apply skills to design business models, calculate IRR/payback, and draft basic PPAs, and present customized customer pitches.

GS3. Perform site surveys and feasibility studies, including shading analysis, mapping, and assessment of risks for solar projects.

GS4. Develop the ability to map suitable business models (CAPEX, OPEX, etc.) to different consumer categories (residential, C&I, government).

GS5. Build professional capacity to work with multi-stakeholder environments (developers, financiers, DISCOMs, regulators) and align solar projects with policy frameworks and grid evacuation requirements

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Business Models in Solar PV	**7**	**6**	**-**	**-**
PC1. Discuss the overview of the Indian solar industry, key stakeholders (developers, EPCs, DISCOMs, financiers), and market drivers.	1	-	-	-
PC2. Discuss different types of business models CAPEX, OPEX/RESCO, Lease, BOOT, and Hybrid models	1	-	-	-
PC3. Explain the structure, ownership, financial responsibility, advantages, and typical users of capex model.	1	-	-	-
PC4. Discuss service-based approach, power purchase agreements (PPA), tariff structures, and stakeholder roles for OPEX model.	1	-	-	-
PC5. Business logic for shared solar, community solar systems, and consumer aggregation	1	-	-	-
PC6. Describe payback periods, IRR, net savings, and other performance indicators.	1	-	-	-
PC7. Discuss the role of MNRE, state policies, subsidies, and DISCOM regulations influencing business models.	1	-	-	-
PC8. Case study analysis of a rooftop solar project using CAPEX model	-	1	-	-
PC9. Design a basic PPA for a RESCO project	-	1	-	-
PC10. Calculate Simple IRR and payback period calculation using Excel	-	1	-	-
PC11. Customer pitch for different business models	-	1	-	-
PC12. Show how to do mapping of solar business models to different consumer categories (residential, C&I, government)	-	1	-	-
PC13. Show the role of MNRE, state policies, subsidies, and DISCOM regulations influencing business models by visiting their website.	-	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Site Feasibility Study Rooftop Solar	**15**	**10**	**-**	**-**
PC14. Identify optimum location of rooftop solar plants installations	1	-	-	-
PC15. Assess the site level pre-requisites for solar panel installation	1	-	-	-
PC16. Discuss how to decide on the type of mounting to be constructed and place of mounting as per client requirement	1	-	-	-
PC17. Discuss how to check for any shading obstacles	1	-	-	-
PC18. Discuss to prepare a site map of the location where installation has to be carried out.	1	-	-	-
PC19. Discuss how to assess the load to be run on solar PV power plant and prepare a load profile	1	-	-	-
PC20. Explain to estimate the capacity of solar PV power plant	1	-	-	-
PC21. Discuss on how to decide on battery backup as per grid availability, loads and client expectation	1	-	-	-
PC22. Assess or obtain the site-specific major parameters of solar resource data like GHI, DNI, Temperature and Wind	1	-	-	-
PC23. Explain how to perform shading analysis	3	-	-	-
PC24. Discuss how to estimate the energy generated from the rooftop solar PV power plant using solar design software like PV Sol, PVsyst, etc.	1	-	-	-
PC25. Identify the risks associated with the specific solar project	1	-	-	-
PC26. Explain how to prepare a site Feasibility Study Report using specialized software like PV sol, PV Syst, etc.	1	-	-	-
PC27. Demonstrate how to identify optimum location of rooftop solar plants installations	-	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Demonstrate how to assess the site level pre-requisites for solar panel installation.	-	1	-	-
PC29. Show how to decide on the type of mounting to be constructed and place of mounting as per client requirement.	-	1	-	-
PC30. Demonstrate how to check for any shading obstacles, perform shading analysis, and to prepare a site map of the location where installation has to be carried out.	-	1	-	-
PC31. Analyze how to assess the load to be run on solar PV power plant, prepare a load profile and estimate the capacity of solar PV power plant	-	1	-	-
PC32. Demonstrate how to decide on battery backup as per grid availability, loads and client expectation	-	1	-	-
PC33. Show how to obtain the site-specific parameters of solar resource data like GHI, DNI, Temperature and Wind	-	1	-	-
PC34. Show how to estimate the energy generated from the rooftop solar PV power plant using solar design software like PV SOL, PVsyst, etc.	-	1	-	-
PC35. Demonstrate the risks associated with the specific solar project and mitigate those	-	1	-	-
PC36. Demonstrate how to support a site Feasibility Study Report using specialized software like PVsyst, etc	-	1	-	-
Site Feasibility Study of Ground Mounted Solar	**8**	**4**	**-**	**-**
PC37. Explain how to assess irradiation and other geological data	1	-	-	-
PC38. Explain how to assess the daily, monthly and annual solar resource data including GHI, ONI, Albedo etc. for site to evaluate the potential for solar energy generation at the site.	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC39. Explain how to collect data of local weather conditions such as temperature range, flooding, wind speed, humidity, pollution levels, snow and other climatic conditions for assessment of its impact on solar energy generation.	1	-	-	-
PC40. Explain how to conduct the soil testing and contour mapping.	1	-	-	-
PC41. Explain how to conduct the detailed survey plan of the land proposed for installation of solar power plant with elevations and topography contour mapping etc.	1	-	-	-
PC42. Discuss how to assess physical site accessibility	1	-	-	-
PC43. Discuss to identify the policy, regulations and procedures for ground mounted solar sector.	1	-	-	-
PC44. Discuss how to assess transmission line & power evacuation	1	-	-	-
PC45. Demonstrate how to retrieve irradiation data from site	-	1	-	-
PC46. Analyse daily, monthly and annual resource data including GHI, DNI, Albedo etc. for site to evaluate the potential for solar energy generation at the site.	-	1	-	-
PC47. Analyse the soil test report	-	1	-	-
PC48. Demonstrate how to use site survey tools like, hand held irradiation meter, compass.	-	1	-	-
NOS Total	**30**	**20**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4125
NOS Name	Business Model and site feasibility study report
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

SGJ/N0148: Manage Installation of Solar Power Plant

Description

This unit is about managing installation, operation and maintenance of Solar Rooftop power plant and Solar Ground Mounted PV Plant

Scope

The scope covers the following :

- Installation of Solar PV rooftop plant
- Operation and maintenance of Solar PV rooftop plant and Solar Ground Mounted PV Plant

Elements and Performance Criteria

Installation of Solar PV rooftop plant

To be competent, the user/individual on the job must be able to:

PC1. explain how to oversee casting of civil foundation on roof

PC2. ensure installation tools availability on site

PC3. oversee the process of installing mounting structure

PC4. discuss installation of foundation for ACDB, Inverter etc

PC5. explain how to install the Energy meter

PC6. explain how to install the AC isolator

PC7. discuss how to install the Earthing and Lighting arrestor on site

PC8. explain use of electrical installation tools

PC9. explain the process of testing and commissioning of equipment in DC and AC side of the plant

PC10. show how to different components and tools used in rooftop solar power plant installation

PC11. show how to oversee installation of solar rooftop solar power plant

PC12. show how to analyse the process of overseeing the installation of various mounting structures

PC13. show how to oversee the testing and commissioning of equipment in DC and AC side of the solar rooftop plant

Installation of Solar Ground Mounted PV Plant

To be competent, the user/individual on the job must be able to:

PC14. Explain how to do marking of footings and distance between two rows

PC15. Explain how to oversee to cast the Civil foundation (Footing/pedestal) on ground

PC16. Ensure the electrical installation tools shall be available on site

PC17. Discuss how to oversee the process of installing Mounting structure for ground mounted solar plant

PC18. Discuss how to install the foundation for ACDB, Inverter and install canopy for outdoor environment etc

PC19. Explain how to install the Energy meter
PC20. Discuss how to install the AC isolator
PC21. Discuss how to install the Earthing and Lighting arrestor on site for ground mounted plant.
PC22. Discuss the know-how of electrical installation tools
PC23. Explain the process of testing and commissioning of equipment in DC and AC side of the plant.
PC24. Show how to do marking of footings with minimum error and distance between two rows .
PC25. Demonstrate different components and tools used in ground mounted solar power plant installation
PC26. Demonstrate how to oversee installation of solar ground mounted solar power plant
PC27. Analyze the process of overseeing the installation of various module mounting structures for ground mounted PV.
PC28. Demonstrate how to oversee the testing and commissioning of equipment in DC and AC side of the solar ground mounted plant.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the process of civil foundation marking and casting (footings/pedestals) for rooftop and ground-mounted solar plants.
KU2. Gain knowledge of mounting structure installation, foundations for ACDB/Inverter, and canopies for outdoor setups.
KU3. Learn the installation requirements of electrical components such as energy meters, AC isolators, earthing, and lightning arrestors.
KU4. Understand the use and importance of electrical installation tools in solar PV projects.
KU5. Acquire knowledge of testing and commissioning procedures for both DC and AC sides of solar power plants.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate how to perform accurate marking of footings and maintain correct row spacing for solar installations.
GS2. Demonstrate the use of different tools and components required in rooftop and ground-mounted solar plant installation.
GS3. Apply skills to oversee and supervise the complete installation process, including mounting structures, foundations, and electrical equipment.
GS4. Analyze and evaluate the installation quality and structural integrity of module mounting systems for both rooftop and ground-mounted PV.
GS5. Demonstrate the ability to oversee testing and commissioning of DC and AC side equipment to ensure safe and reliable plant operation.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Installation of Solar PV rooftop plant	**19**	**16**	**-**	**-**
PC1. explain how to oversee casting of civil foundation on roof	2	-	-	-
PC2. ensure installation tools availability on site	2	-	-	-
PC3. oversee the process of installing mounting structure	2	-	-	-
PC4. discuss installation of foundation for ACDB, Inverter etc	2	-	-	-
PC5. explain how to install the Energy meter	3	-	-	-
PC6. explain how to install the AC isolator	2	-	-	-
PC7. discuss how to install the Earthing and Lighting arrestor on site	2	-	-	-
PC8. explain use of electrical installation tools	2	-	-	-
PC9. explain the process of testing and commissioning of equipment in DC and AC side of the plant	2	-	-	-
PC10. show how to different components and tools used in rooftop solar power plant installation	-	4	-	-
PC11. show how to oversee installation of solar rooftop solar power plant	-	4	-	-
PC12. show how to analyse the process of overseeing the installation of various mounting structures	-	4	-	-
PC13. show how to oversee the testing and commissioning of equipment in DC and AC side of the solar rooftop plant	-	4	-	-
Installation of Solar Ground Mounted PV Plant	**10**	**5**	**-**	**-**
PC14. Explain how to do marking of footings and distance between two rows	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Explain how to oversee to cast the Civil foundation (Footing/pedestal) on ground	-	-	-	-
PC16. Ensure the electrical installation tools shall be available on site	1	-	-	-
PC17. Discuss how to oversee the process of installing Mounting structure for ground mounted solar plant	2	-	-	-
PC18. Discuss how to install the foundation for ACDB, Inverter and install canopy for outdoor environment etc	1	-	-	-
PC19. Explain how to install the Energy meter	1	-	-	-
PC20. Discuss how to install the AC isolator	1	-	-	-
PC21. Discuss how to install the Earthing and Lighting arrestor on site for ground mounted plant.	1	-	-	-
PC22. Discuss the know-how of electrical installation tools	1	-	-	-
PC23. Explain the process of testing and commissioning of equipment in DC and AC side of the plant.	1	-	-	-
PC24. Show how to do marking of footings with minimum error and distance between two rows .	-	1	-	-
PC25. Demonstrate different components and tools used in ground mounted solar power plant installation	-	1	-	-
PC26. Demonstrate how to oversee installation of solar ground mounted solar power plant	-	1	-	-
PC27. Analyze the process of overseeing the installation of various module mounting structures for ground mounted PV.	-	1	-	-
PC28. Demonstrate how to oversee the testing and commissioning of equipment in DC and AC side of the solar ground mounted plant.	-	1	-	-
NOS Total	**29**	**21**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0148
NOS Name	Manage Installation of Solar Power Plant
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Installation, Operation and Maintenance
NSQF Level	4
Credits	3
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

SGJ/N4126: Operation and maintenance of Solar Rooftop, Ground Mounted PV Plant

Description

Covers the procedures for operating, inspecting, troubleshooting, and maintaining both rooftop and ground-mounted solar PV systems to ensure safe, reliable, and efficient performance.

Scope

The scope covers the following :

- Includes monitoring energy output, cleaning modules, checking electrical/electronic equipment, vegetation and soil control, grid interface monitoring, and carrying out preventive and corrective maintenance activities.

Elements and Performance Criteria

Operation and maintenance of Solar Rooftop PV Plant

To be competent, the user/individual on the job must be able to:

PC1. Explain how to handle and utilize tools and equipment for O&M available at site.
PC2. Explain how to check the integrity and fuses and circuit breakers.
PC3. Ensure working condition of the junction boxes/combiner boxes.
PC4. Ensure the continuity of cables and wires are maintained.
PC5. Ensure proper electrical connections throughout the solar PV power plant up to the inverter input.
PC6. Discuss how to verify the connections, cables and junction boxes as per the design/ working drawings.
PC7. Explain how to troubleshoot the identified faults and escalate the issue to concerned person for timely resolution.
PC8. Explain how to verify the earthing and lightening protection systems as per the as-built drawings and report in case of any discrepancies.
PC9. Ensure proper cleaning of modules as per schedule and standard procedure and remove any shadowing objects.
PC10. Discuss how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance.
PC11. Ensure that the system is well maintained and is performing at an optimum level.
PC12. Discuss best practices and successful case studies on performing O&M.
PC13. Demonstrate how to handle tools and equipment for O&M available at site.
PC14. Demonstrate how to check the integrity and working condition of all connections, fuses, circuit breakers and all other components of the rooftop solar power plant.
PC15. Discuss how to verify the connections, cables and junction boxes as per the design/ working drawings.

PC16. Demonstrate how to ensure that the system is well maintained and is performing at an optimum level.
PC17. Demonstrate how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance.
PC18. Show how to interpret a successful case study on operation and maintenance of solar rooftop power plant.
PC19. Demonstrate best practices on O&M as employed by the industry

Operation and maintenance of Solar Ground Mounted PV Plant

To be competent, the user/individual on the job must be able to:

PC20. Discuss the Purpose, scope, and importance of operation and maintenance in ground mounted solar plants.
PC21. Define different types of maintenance with examples, schedules, and benefits.
PC22. Discuss the tools and equipment for O&M available at site and how to check the integrity and fuses and circuit breakers.
PC23. Ensure working condition of the junction boxes/combiner boxes.
PC24. Ensure the continuity of cables and wires are maintained.
PC25. Ensure proper electrical connections throughout the solar PV power plant up to the inverter input.
PC26. Discuss how to verify the connections, cables and junction boxes as per the design/ working drawings.
PC27. Explain how to troubleshoot the identified faults and escalate the issue to concerned person for timely resolution.
PC28. Discuss to identify the faults in cables.
PC29. Explain how to verify the earthing and lightening protection systems as per the as-built drawings and report in case of any discrepancies.
PC30. Ensure proper cleaning of modules as per schedule and standard procedure and remove any shadowing objects.
PC31. Discuss how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance.
PC32. Ensure that the system is well maintained and is performing at an optimum level.
PC33. Discuss best practices and successful case studies on performing O&M.
PC34. Show the Purpose, scope, and importance of operation and maintenance in ground mounted solar plants with some videos and examples.
PC35. Illustrate how to identify the type of maintenance required for a plant
PC36. Show how to use tools and equipment for O&M available at site.
PC37. Demonstrate how to check the integrity and working condition of all connections, fuses, circuit breakers and all other components of the ground mounted solar power plant.
PC38. Show how to verify the connections, cables and junction boxes as per the design/ working drawings of ground mounted plants.
PC39. Demonstrate how to ensure that the system is well maintained and is performing at an optimum level.
PC40. Demonstrate how to do maintenance of cables and conduits

PC41. Demonstrate how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance for ground mounted plants.

PC42. Show how to interpret a successful case study on operation and maintenance of solar rooftop power plant

PC43. Demonstrate best practices on O&M as employed by the industry

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the purpose, scope, and importance of O&M in rooftop and ground-mounted solar PV plants, including its role in ensuring long-term performance.

KU2. Gain knowledge of types of maintenance (scheduled/preventive and unscheduled/corrective) with their examples, schedules, and benefits.

KU3. Learn the process of checking and verifying system components (connections, fuses, breakers, junction boxes, cables, inverters) as per design and as-built drawings.

KU4. Understand how to troubleshoot and escalate faults, and verify the effectiveness of earthing and lightning protection systems.

KU5. Acquire knowledge of best practices, case studies, and industry standards in module cleaning, electrical checks, and performance monitoring for optimum plant efficiency.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate how to handle and utilize O&M tools and equipment safely and effectively at site.

GS2. Apply skills to check the integrity and working condition of all connections, fuses, circuit breakers, junction boxes, and cables.

GS3. Demonstrate how to verify connections and perform routine inspections to ensure system performance is as per design specifications.

GS4. Implement scheduled and corrective maintenance approaches, including cleaning, cable/conduit maintenance, and system optimization.

GS5. Demonstrate the ability to analyze and interpret O&M case studies, adopt industry best practices, and ensure professional reporting of system health.

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Operation and maintenance of Solar Rooftop PV Plant	**12**	**9**	**-**	**-**
PC1. Explain how to handle and utilize tools and equipment for O&M available at site.	1	-	-	-
PC2. Explain how to check the integrity and fuses and circuit breakers.	1	-	-	-
PC3. Ensure working condition of the junction boxes/combiner boxes.	1	-	-	-
PC4. Ensure the continuity of cables and wires are maintained.	1	-	-	-
PC5. Ensure proper electrical connections throughout the solar PV power plant up to the inverter input.	1	-	-	-
PC6. Discuss how to verify the connections, cables and junction boxes as per the design/ working drawings.	1	-	-	-
PC7. Explain how to troubleshoot the identified faults and escalate the issue to concerned person for timely resolution.	1	-	-	-
PC8. Explain how to verify the earthing and lightening protection systems as per the as-built drawings and report in case of any discrepancies.	1	-	-	-
PC9. Ensure proper cleaning of modules as per schedule and standard procedure and remove any shadowing objects.	1	-	-	-
PC10. Discuss how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance.	1	-	-	-
PC11. Ensure that the system is well maintained and is performing at an optimum level.	1	-	-	-
PC12. Discuss best practices and successful case studies on performing O&M.	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Demonstrate how to handle tools and equipment for O&M available at site.	-	1	-	-
PC14. Demonstrate how to check the integrity and working condition of all connections, fuses, circuit breakers and all other components of the rooftop solar power plant.	-	2	-	-
PC15. Discuss how to verify the connections, cables and junction boxes as per the design/ working drawings.	-	2	-	-
PC16. Demonstrate how to ensure that the system is well maintained and is performing at an optimum level.	-	1	-	-
PC17. Demonstrate how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance.	-	1	-	-
PC18. Show how to interpret a successful case study on operation and maintenance of solar rooftop power plant.	-	1	-	-
PC19. Demonstrate best practices on O&M as employed by the industry	-	1	-	-
Operation and maintenance of Solar Ground Mounted PV Plant	**16**	**13**	**-**	**-**
PC20. Discuss the Purpose, scope, and importance of operation and maintenance in ground mounted solar plants.	1	-	-	-
PC21. Define different types of maintenance with examples, schedules, and benefits.	1	-	-	-
PC22. Discuss the tools and equipment for O&M available at site and how to check the integrity and fuses and circuit breakers.	2	-	-	-
PC23. Ensure working condition of the junction boxes/combiner boxes.	1	-	-	-
PC24. Ensure the continuity of cables and wires are maintained.	1	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. Ensure proper electrical connections throughout the solar PV power plant up to the inverter input.	1	-	-	-
PC26. Discuss how to verify the connections, cables and junction boxes as per the design/ working drawings.	1	-	-	-
PC27. Explain how to troubleshoot the identified faults and escalate the issue to concerned person for timely resolution.	1	-	-	-
PC28. Discuss to identify the faults in cables.	1	-	-	-
PC29. Explain how to verify the earthing and lightening protection systems as per the as-built drawings and report in case of any discrepancies.	2	-	-	-
PC30. Ensure proper cleaning of modules as per schedule and standard procedure and remove any shadowing objects.	1	-	-	-
PC31. Discuss how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance.	1	-	-	-
PC32. Ensure that the system is well maintained and is performing at an optimum level.	1	-	-	-
PC33. Discuss best practices and successful case studies on performing O&M.	1	-	-	-
PC34. Show the Purpose, scope, and importance of operation and maintenance in ground mounted solar plants with some videos and examples.	-	2	-	-
PC35. Illustrate how to identify the type of maintenance required for a plant	-	1	-	-
PC36. Show how to use tools and equipment for O&M available at site.	-	1	-	-
PC37. Demonstrate how to check the integrity and working condition of all connections, fuses, circuit breakers and all other components of the ground mounted solar power plant.	-	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC38. Show how to verify the connections, cables and junction boxes as per the design/ working drawings of ground mounted plants.	-	2	-	-
PC39. Demonstrate how to ensure that the system is well maintained and is performing at an optimum level.	-	1	-	-
PC40. Demonstrate how to do maintenance of cables and conduits	-	1	-	-
PC41. Demonstrate how to implement approaches for O&M including scheduled/preventive and unscheduled or corrective maintenance for ground mounted plants.	-	1	-	-
PC42. Show how to interpret a successful case study on operation and maintenance of solar rooftop power plant	-	1	-	-
PC43. Demonstrate best practices on O&M as employed by the industry	-	2	-	-
NOS Total	**28**	**22**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4126
NOS Name	Operation and maintenance of Solar Rooftop, Ground Mounted PV Plant
Sector	Green Jobs
Sub-Sector	
Occupation	Entrepreneur
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

SGJ/N0106: Maintain Personal Health & Safety at project site

Description

This unit is about maintaining health, safety and hygiene at workplace.

Scope

The scope covers the following :

- Adopt safe practices at workplace
- Follow emergencies, rescue and first aid procedures
- Follow good housekeeping practices and infection control guidelines

Elements and Performance Criteria

Adopt safe practices at workplace

To be competent, the user/individual on the job must be able to:

PC1. explain the requirements for safe work area

PC2. identify and report any hazards, risks or breaches in site safety to the appropriate authority

PC3. follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk

PC4. use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition

PC5. follow safe practices when working at height and in confined space

PC6. handle all required tools, tackles, materials and equipment safely

PC7. identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures

PC8. apply ergonomic principles wherever required

PC9. use safety signs, labels, charts and notices at workplace

PC10. identify work safety procedures and instructions for handling heavy components

Follow emergencies, rescue and first aid procedures

To be competent, the user/individual on the job must be able to:

PC11. follow emergency and evacuation procedures in case of accidents, fires and natural calamities

PC12. use appropriate fire extinguishers for different types of fire

PC13. administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.

PC14. use correct method to move injured person during an emergency

Follow good housekeeping practices and infection control guidelines

To be competent, the user/individual on the job must be able to:

PC15. follow recommended personal hygiene, workplace hygiene and sanitation practices

PC16. clean and disinfect all material, tools and supplies before and after use

PC17. report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues

PC18. follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. importance of safety drills

KU2. importance of working in clean and safe environment

KU3. health and safety roles and responsibilities of relevant personnel within and outside the organization

KU4. reporting procedures in case of breaches or hazards for site safety, accidents and emergency situations

KU5. basic ergonomic principle

KU6. key internal and external source of health and safety information

KU7. meaning of hazards, risk and near miss

KU8. importance of Personal Protective Equipment required for specific job

KU9. forms and classification of hazardous substances

KU10. health effect associated with exposure to environmental pollution

KU11. housekeeping activities relevant to task

KU12. symptoms of infection like fever, cough, swelling and inflammation

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. record data on waste disposal at workplace

GS2. complete statutory documents relevant to safety and hygiene

GS3. fill safety formats for near miss, unsafe condition

GS4. identify potential safety risk and report to appropriate authority

GS5. communicate and collaborate with others to incorporate sustainable practices

GS6. interpret general safety guidelines, labels, charts and signage

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Adopt safe practices at workplace	**13**	**19**	**-**	**-**
PC1. explain the requirements for safe work area	2	-	-	-
PC2. identify and report any hazards, risks or breaches in site safety to the appropriate authority	2	3	-	-
PC3. follow recommended safe practices in handling physical, chemical, electrical and fire hazards and risk	1	2	-	-
PC4. use appropriate Personal Protective Equipment(PPE) for head, eye, hand, ear, face, body and fall protection specific to work condition	2	4	-	-
PC5. follow safe practices when working at height and in confined space	1	1	-	-
PC6. handle all required tools, tackles, materials and equipment safely	1	2	-	-
PC7. identify expiry dates, wear and tear issues of specified equipment and accordingly inform supervisor and undertake corrective measures	1	2	-	-
PC8. apply ergonomic principles wherever required	1	2	-	-
PC9. use safety signs, labels, charts and notices at workplace	1	1	-	-
PC10. identify work safety procedures and instructions for handling heavy components	1	2	-	-
Follow emergencies, rescue and first aid procedures	**4**	**4**	**-**	**-**
PC11. follow emergency and evacuation procedures in case of accidents, fires and natural calamities	1	1	-	-
PC12. use appropriate fire extinguishers for different types of fire	1	1	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. administer first aid to victim in case of various medical emergencies including bleeding, burns, choking, electric shock, cardiac arrest, etc.	1	1	-	-
PC14. use correct method to move injured person during an emergency	1	1	-	-
Follow good housekeeping practices and infection control guidelines	**4**	**6**	**-**	**-**
PC15. follow recommended personal hygiene, workplace hygiene and sanitation practices	1	1	-	-
PC16. clean and disinfect all material, tools and supplies before and after use	1	1	-	-
PC17. report immediately to concerned authorities regarding sign and symptoms of illness of self and other colleagues	1	2	-	-
PC18. follow processes specified for disposal of hazardous waste	1	2	-	-
NOS Total	**21**	**29**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0106
NOS Name	Maintain Personal Health & Safety at project site
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health & Safety , Installation, Solar Panel Installation Technician
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	08/05/2025
Next Review Date	07/05/2028
NSQC Clearance Date	08/05/2025

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. identify employability skills required for jobs in various industries

PC2. identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC4. follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC5. recognize the significance of 21st Century Skills for employment

PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC7. use basic English for everyday conversation in different contexts, in person and over the telephone

PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC9. write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC10. understand the difference between job and career

PC11. prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings

PC13. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC14. communicate and behave appropriately with all genders and PwD

PC15. escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC16. select financial institutions, products and services as per requirement

PC17. carry out offline and online financial transactions, safely and securely

PC18. identify common components of salary and compute income, expenses, taxes, investments etc

PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC20. operate digital devices and carry out basic internet operations securely and safely

PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively

PC22. use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research

PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC26. identify different types of customers

PC27. identify and respond to customer requests and needs in a professional manner.

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all
GS4. how to work in a virtual mode
GS5. perform calculations efficiently
GS6. solve problems effectively
GS7. pay attention to details
GS8. manage time efficiently
GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
Constitutional values – Citizenship	**1**	**1**	**-**	**-**
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**4**	**-**	**-**
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
Communication Skills	**2**	**2**	**-**	**-**
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**3**	**4**	**-**	**-**
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Entrepreneurship	**2**	**3**	**-**	**-**
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**3**	**-**	**-**
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.

2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.

3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.

4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of % aggregate marks to successfully clear the assessment.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N0111.Set up New Venture for Solar Photovoltaic System	82	18	0	0	100	25
SGJ/N4125.Business Model and site feasibility study report	30	20	0	0	50	20
SGJ/N0148.Manage Installation of Solar Power Plant	29	21	0	0	50	20
SGJ/N4126.Operation and maintenance of Solar Rooftop, Ground Mounted PV Plant	28	22	0	0	50	15
SGJ/N0106.Maintain Personal Health & Safety at project site	21	29	-	-	50	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	**210**	**140**	**-**	**-**	**350**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.